

# Human-in-the-Loop Gaussian Model Enhancement with Mobile Robotic Re-Capture

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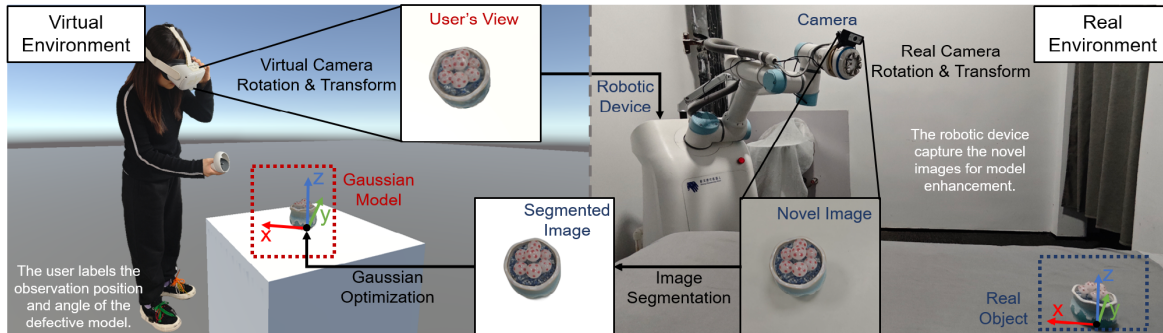


Figure 1: Users can label the viewpoints of the Gaussian model with poor visual quality in the virtual environment. We record the transformation and rotation of the user's head-mounted display in the object coordinate system. In the real environment, the robotic arm with a camera is manipulated to align with the corresponding positions to re-capture the images, which are segmented and subsequently used to enhance the Gaussian model.

## ABSTRACT

3D Gaussian Splatting can create virtual models with captured real-world images using a camera mounted on a mobile robotic arm. However, the Gaussian model quality deteriorates when parts of the object that have not been previously captured are exposed in VR interaction. This poster presents a human-in-the-loop Gaussian model enhancement method, comprising pre-training, viewpoint labeling, robotic image re-capture, and Gaussian model enhancement. By incorporating newly re-captured images, the enhanced models achieve improved image similarity metrics compared to the pre-trained ones, with minimal optimization time. This method facilitates continuous quality improvement, making Gaussian models more adaptable for VR applications.

**Index Terms:** Computing methodologies Computer graphics→ Rendering; Point-based models.

## 1 INTRODUCTION

3D Gaussian Splatting is a novel rendering technique that represents virtual objects using multiple anisotropic 3D Gaussians, enabling efficient rendering by rasterizing these Gaussians into images [1]. This method facilitates rapid model generation from multi-view images captured in real-world settings, offering new possibilities for virtual model generation in virtual reality (VR). Despite its high efficiency advantages, this technology is challenged by the potential for user interaction with objects in VR, such as moving, rotating, or deforming objects (e.g., opening the lid of a box), which can expose parts of the object that have not previously been captured, leading to a suboptimal visual quality of the Gaussian model [2]. While methods like VR-Splatting have proposed scanning a large number of images and employing foveated rendering as a potential solution [3], this approach also requires a significant amount of time for Gaussian optimization. With advances in mobile robotic technologies, devices such as robotic arms, drones, and other camera-mounted systems

have emerged as a dominant approach to scanning real-world objects. By capturing images from novel viewpoints using mobile robotic devices and utilizing them for Gaussian optimization, the visual quality of the Gaussian model can be effectively enhanced with minimal optimization time.

This poster presents a human-in-the-loop Gaussian model enhancement method, as shown in Fig. 1. In the real environment, a mobile robotic arm first captures images from predefined positions and generates a pre-trained Gaussian model. Subsequently, the user observes the Gaussian model in the virtual environment (VE) and labels viewpoints that exhibit poor visual quality in VR interaction. These labeled transformations and rotations are then synchronized with the real-world robotic arm, which re-captures images of the object from the corresponding viewpoints. The newly captured images are segmented and utilized for Gaussian optimization, enhancing the model's visual quality.

## 2 METHOD AND SYSTEM IMPLEMENTATION

### 2.1 Gaussian Model Pre-training

Our first step involves generating a pre-trained Gaussian model using a camera mounted on the mobile robotic arm. We developed a prototype system with a computer featuring an Intel i7 processor and an NVIDIA 3080 GPU for rendering. The system utilized a Shining Robot [4], with a Logitech C930 camera mounted on the terminal of the robotic arm. The camera captured images at a resolution of  $720 \times 720$ , and both the robotic arm and camera were connected to the computer for data processing and synchronization. The robotic arm's reachable range exceeded a sphere with a 30cm radius, and the terminal's movement speed was set to 5cm per second to ensure synchronized image capture. In the real environment, we established the object coordinate system by defining the center of the bottom of the potted plant as the origin, with the x-axis pointing directly in front and the z-axis pointing upward. The robotic arm was programmed to drive the camera to scan the potted plant horizontally from top to bottom along a 25cm radius, taking 24 images at 45 degree intervals. The transformation and rotation parameters of the

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camera corresponding to these images were recorded in the object coordinate system. Next, we employed the 3D Gaussian Splatting to pre-train the Gaussian model.

## 2.2 Human-in-the-Loop Viewpoint Labeling

The pre-trained Gaussian model may exhibit poor visual quality when parts of the object that have not been previously captured are exposed in VR interaction. To address this, we incorporated human participants to label these viewing transformations and rotations. The Gaussian model was imported into a Unity Gaussian Splatting project [5], where users, wearing a Meta Quest 2 head-mounted display (HMD), observed the model and used the controller buttons to label the viewpoints with poor visual quality. At each viewpoint, we recorded the transform matrix  $T \in \mathbb{R}^{3 \times 4}$  and the rotation matrix  $R \in \mathbb{R}^{3 \times 3}$  of the HMD's camera in the virtual world coordinate system. We defined the object coordinate system in the VE with the center of the Gaussian model as the origin, the forward direction as the x-axis, and the upward direction as the z-axis. This virtual object coordinate system was aligned with the corresponding real object coordinate system. Next, we changed the HMD camera's transformations and rotations from the world coordinate system into the object coordinate system by calculating the transform matrix  $T^o$  and rotation matrix  $R^o$  in the object coordinate system.

## 2.3 Mobile Robotic Image Re-capture

The viewpoints labeled can be synchronized with the real-world robotic arm to re-capture images. In the real environment's object coordinate system, the transformation for the real camera is derived from  $T^o$ , and the rotation is derived from  $R^o$ . The robotic arm is then moved, driving the camera to the specified position to capture an image from the viewpoint where the user labeled poor visual quality in the VE. Afterward, the captured image is automatically processed by Rembg [6] to segment the background, retaining only the potted plant for further processing.

## 2.4 Gaussian model enhancement

The segmented images and corresponding camera parameters are used to enhance the Gaussian model. We leverage the Gaussian optimization with a key distinction from the pre-training phase: instead of re-aligning multi-view images and generating SfM points, we augment the existing Gaussian model using only newly captured images. In other words, we initialize the optimization process with the pre-trained Gaussian model parameters and then perform optimization based solely on the new images and corresponding camera parameters. This approach accelerates the Gaussian optimization process, aiming to minimize the delay associated with the human-in-the-loop enhancement procedure. After model enhancement, the updated Gaussian model is re-introduced into the VE. In VR interactions, the user can continue to observe and label viewpoints, facilitating an iterative process of continuous enhancement.

## 3 EVALUATION

We conducted proof-of-concept experiments on the system described in Section 2. Initially, the robotic arm captured images at 24 predefined viewpoints to pre-train and generate the Gaussian model. Following this, the user labeled the viewpoints where he felt the model quality was poor, and the robotic arm was synchronized to those positions to re-capture images from the identified viewpoints. During the Gaussian model enhancement, we initialized the model using the pre-trained Gaussian parameters and enhanced it by incorporating different numbers of newly captured images. The optimization process used a combination of  $L_1$  loss and the structural similarity index measure loss, defined as  $L = 0.8 \times L_1 + 0.2 \times L_{D-SSIM}$ . In the experiment, the user labeled one to four viewpoints. We evaluated the model quality using image similarity metrics under the viewpoints, including PSNR, SSIM, and LPIPS. Additionally, we recorded the

Table 1: Image similarity metrics and optimization times for the pre-trained and enhanced Gaussian models.

| Model              | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | Opt Time |
|--------------------|-----------------|-----------------|--------------------|----------|
| Pre-trained Model  | 23.56           | 0.907           | 0.137              | 948 s    |
| +1 Captured Image  | 25.71           | 0.920           | 0.130              | 26 s     |
| +2 Captured Images | 26.69           | 0.924           | 0.126              | 42 s     |
| +3 Captured Images | 27.28           | 0.937           | 0.121              | 52 s     |
| +4 Captured Images | 27.86           | 0.958           | 0.119              | 64 s     |

time taken for the loss  $L$  to stabilize. Table 1 presents the image similarity metrics and optimization times for the pre-trained and enhanced Gaussian models after incorporating different numbers of newly captured images.

The results demonstrate that incorporating re-captured images improves the quality of the Gaussian model in those viewpoints compared to the original pre-trained Gaussian model. Additionally, introducing multiple newly captured images further enhances the image similarity metrics. The optimization time required to refine the pre-trained Gaussian model with newly captured images is relatively short. Through human-in-the-loop observation and labeling, the visual quality of the Gaussian model can be continuously improved in VR interaction. This iterative process ensures the Gaussian model becomes increasingly suitable for integration into VR applications.

## 4 DISCUSSION AND FUTURE WORK

The Gaussian splatting technique enables the rapid generation of virtual models from real multi-view images. However, these images may not capture all viewpoints of the real model, which can lead to poor quality in the Gaussian model when parts of the object that have not been previously captured are exposed in VR interaction. This paper proposes a human-in-the-loop Gaussian model enhancement method to address this issue and quickly improve the visual quality of the Gaussian model. Our proof-of-concept experiments demonstrate that users can observe and label viewpoints with poor visual quality in the Gaussian model within the VE. The robotic arm then drives the camera synchronizing to these labeled viewpoints, allowing for the re-capturing of images that enhance the model's quality. We encourage researchers to expand on this method by employing more cameras than a single camera and using robots with a broader range of motion, such as drones. Future work could also involve increasing the number of participants in the human-in-the-loop labeling process to generate larger datasets for assessing Gaussian model quality in VR, facilitating automated Gaussian model enhancement.

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## REFERENCES

- [1] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4):1–14, 2023.
- [2] Ben Fei, Jingyi Xu, Rui Zhang, Qingyuan Zhou, Weidong Yang, and Ying He. 3d gaussian splatting as new era: A survey. *IEEE Transactions on Visualization and Computer Graphics*, 2024.
- [3] Linus Franke, Laura Fink, and Marc Stamminger. Vr-splatting: Foveated radiance field rendering via 3d gaussian splatting and neural points. *arXiv preprint arXiv:2410.17932*, 2024.
- [4] Limber robot. <https://www.limberrobot.com.cn/>, 2024.
- [5] Aras Pranckevičius. Unity gaussian splatting. <https://github.com/aras-p/UnityGaussianSplatting>, 2024.
- [6] Daniel Gatis. Rembg. <https://github.com/danielgatis/rembg>, 2024.